# 1. Identity

| | |
|---|---|
| Document | STAR-PKG-STAR-COORDS-EN · v2.2 · 2026-11-02 |
| Package | star-coords |
| Version | 0.1.0 |
| Capability | star.geometry.coords |
| Maturity | Released & Verified (RELEASED) |
| Python | ⩾3.9 |
| Source commit | dc09f61 |
| Source | https://github.com/Andrea07072000/S.T.A.R.-/tree/main/packages/star_coords |

# 2. Licence and obligations

Apache License 2.0. You may use, copy, modify and distribute it, also commercially, and you receive a patent licence from the contributors. When you redistribute it: keep the LICENSE file and the copyright notices (and a NOTICE file, if a future version adds one), and state in modified files that you changed them. You may not use the S.T.A.R. name to endorse your product.

# 3. What it needs at run time

Nothing beyond the Python standard library: it works offline.

# 4. Integrity

SHA-256 fingerprints of the files published on this website. Check a downloaded file with:

```
python -c "import hashlib,sys;print(hashlib.sha256(open(sys.argv[1],'rb').read()).hexdigest())" star_coords-0.1.0-py3-none-any.whl
```

| FILE | SHA-256 |
|---|---|
| star_coords-0.1.0-py3-none-any.whl | b07cba1f91bc503c2841761dc1277f0e8f7bcce5077a01fc5f502eb5305fd02e |
| star_coords.py | 80c6c016a17f7b4d2066eb5eba4708ccae6f151333e346a052f990b1249b80d0 |
| star_coords-0.1.0-source.zip | 84f26c0a450caa639295ee7a38c081dddafa08db520da9a810b7b2a3ed06438a |

# 5. What it does, and its limits

Spherical and cylindrical coordinates to rectangular and back; agrees with ERFA, SPICE and astropy to 3e-16 relative and 6e-14 deg on 600 cases in both directions, close to the axis included, and refuses out-of-range input.

Geocentric (spherical) latitude, not geodetic: for the ellipsoid use `star_geodesy`. No velocities. Colatitude conventions (physics spherical coordinates) are not offered: latitude is from the xy plane.

- `spherical_to_xyz(lon, lat, r)` and `xyz_to_spherical(x, y, z)` follow x = r cos(lat) cos(lon), y = r cos(lat) sin(lon), z = r sin(lat); longitude out in [0, 360), latitude from the arc-tangent (accurate at the poles), length without intermediate overflow or underflow.
- `cylindrical_to_xyz(rho, lon, z)` and `xyz_to_cylindrical(x, y, z)`: x = rho cos(lon), y = rho sin(lon).
- Measured on 600 cases (100 directions down to 1e-9 deg from the z axis, lengths from 1e-3 to 1e9), in both directions: components and lengths within 3e-16 relative and angles within 6e-14 deg of displacement of ERFA, SPICE and astropy (spherical) and of SPICE and astropy (cylindrical).
- Every function returns finite floats or raises `ValueError`: non-numeric, boolean, NaN or infinite input, longitude outside [-360, 360], latitude outside [-90, 90], a negative radius, a length or component above 1e300. On the z axis the longitude is undefined and returned as 0.

# 6. Export, use and certification

The package is published and freely available. That usually takes software out of export-control lists, but not every use of it: read the page on space activities and export control before

using it in a controlled programme. It is not certified or qualified for flight; the evidence describes what was tested and how.

## 7. Citation and bill of materials

```
@software{cavazzini_star_coords_0_1_0,
  author  = {Cavazzini, Andrea},
  title   = {{star-coords}: Spherical and cylindrical ⇄ rectangular coordinates},
  version = {0.1.0},
  year    = {2026},
  url     = {https://andrea07072000.github.io/S.T.A.R.-/software/star_coords/},
  note    = {Apache-2.0. Source: https://github.com/Andrea07072000/S.T.A.R.-, commit dc09f61}
}
```

CITATION.cff · BibTeX · SBOM CycloneDX 1.5 · SBOM SPDX 2.3

## 8. Warranty and liability

Provided "as is", without warranties or conditions of any kind (Apache License 2.0, sections 7 and 8), to the extent the law allows. This never excludes liability that the law does not allow to exclude.

Related: space activities and export control · acceptable use · evidence and limits · licence texts